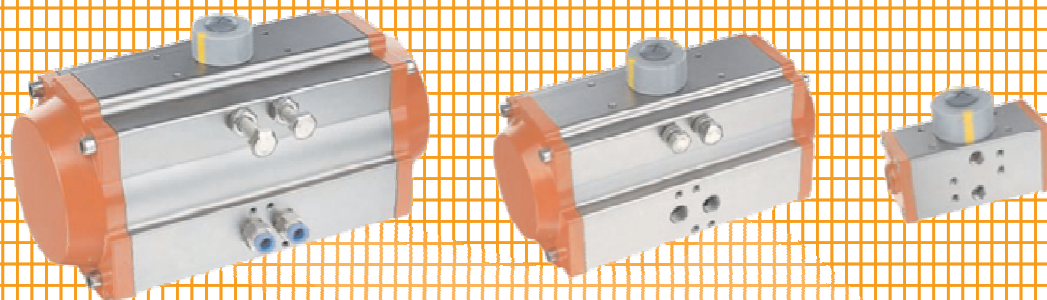


TORQUEMATE

Efficient Solutions for Valve Automation



ROBUST
RELIABLE
RAPID

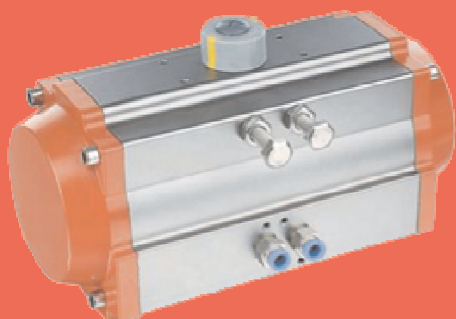
SUITABLE TO ANY 90° ROTATION VALVE.
CAN BE MADE IN DIFFERENT
CONFIGURATIONS FOR ANY APPLICATION
AND ADAPTATION FOR OTHER AUTOMATION
DEVICES.

Advantages:

- ▲ Superior protection on interior and exterior housing surfaces.
- ▲ Actuators meet the latest NAMUR standards, allowing direct or close coupling of NAMUR designed accessories.
- ▲ Modular spring packs simplify the conversion from double-acting to spring-return.

TORQUEMATE PNEUMATIC ACTUATORS

WWW.DANARES.COM



The **TORQUEMATE** Pneumatic “Rack and Pinion” actuator is one of the best designs in the market. Available as “Double Acting” or “Spring Return” with torque outputs for small to large valves.

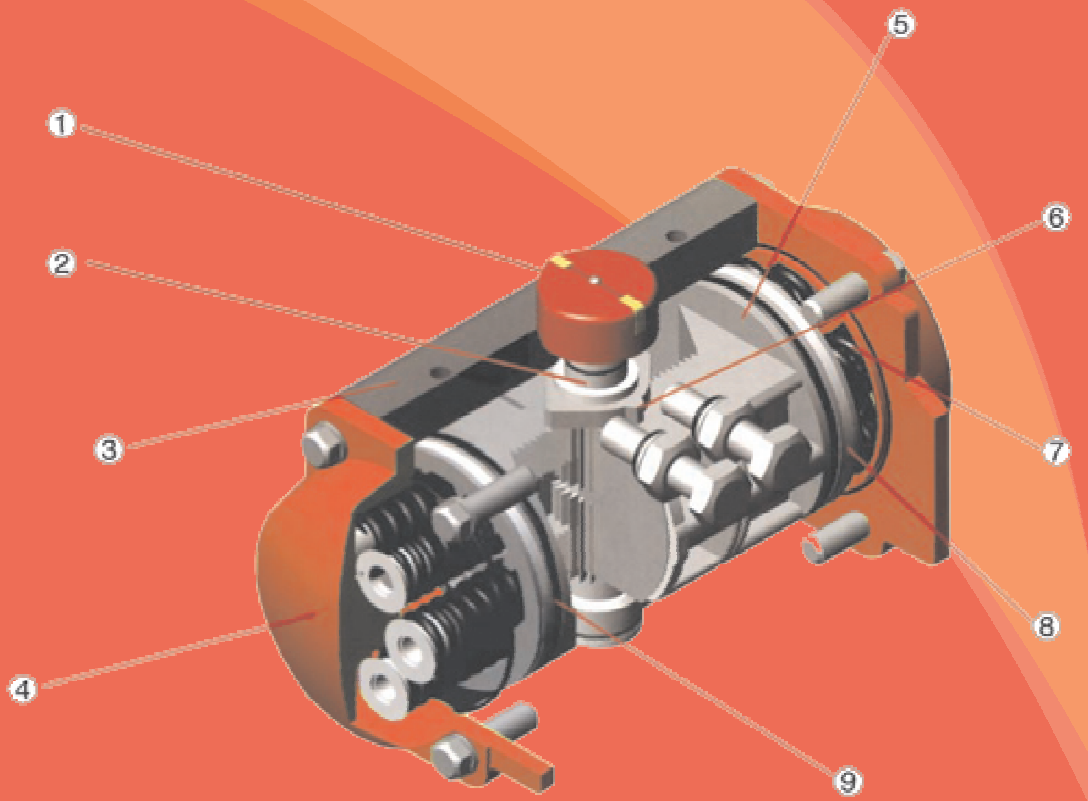
Both actuators have twin cylinders, horizontally opposed incorporated piston guides to assure the correct contact between the “Rack and Pinion”. And transforms the linear motion of it's pneumatic pistons in to rotary motion via two racks that drive the central pinion.

These actuators are prepared with bores and threads that meet the international standards NAMUR and ISO. Built in robust extruded aluminum, but light and resistant to environmental corrosion.

We offer double acting and spring return units with torque outputs for small to larger flanged valves.

Mounting pads are drilled and tapped to international NAMUR and ISO standards. **TORQUEMATE** offers a full array of solenoids, limit switches, positioners and other accessories to meet the customer's needs.

Parts List:



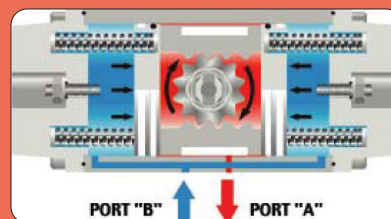
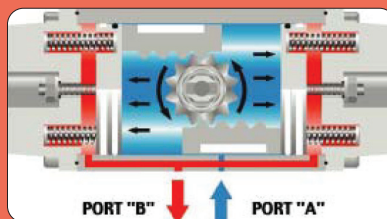
Sl. No.	Description	Qty.	Material
1	Indicator	1	ABS (Plastic)
2	Pinion	1	Steel
3	Actuator Body	1	Extruded Aluminium
4	End Caps	1	AL356/380
5	Pistons	1	AL356/380
6	Travel Adjustment	1	Stainless Steel
7	High performance springs	1	Spring Steel
8	Bearings & Guides	1	Delrin
9	O-rings	1	NBR / EPDM / Viton

Principle of Operation of The Actuator - ISO5211 with spring return.

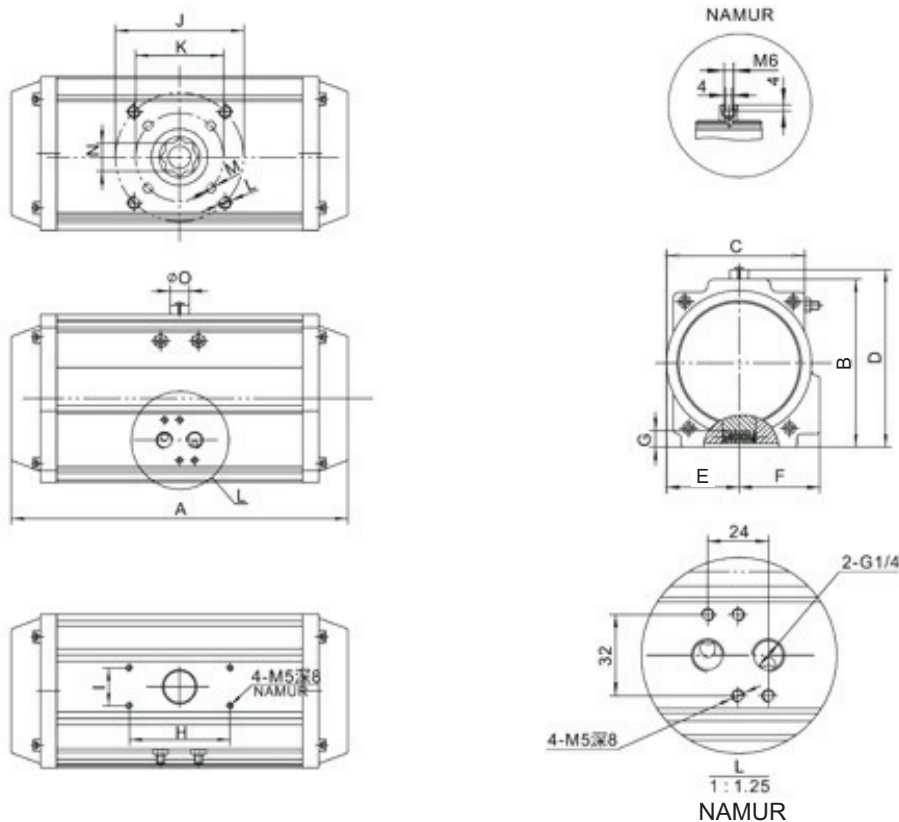
Air pressure applied to the Port "A" will cause the internal cavities to pressurize, forcing the pistons to move toward the end, compressing the springs. The axis rotates in a counter clockwise motion. When stopping the application of air pressure in port "A" the springs will exercise pressure to close the pistons and to rotate the axis quickly in a clockwise motion. This action is frequently used to close the valve referred to as "NORMALLY CLOSED."

The axis will rotate in counter clockwise motion when pressure is applied in Port "A", it forces the pistons to separate.

During this operation, the air of the external cavity is expelled through port "B." The turn of the axis in a clockwise motion is achieved with the action opposite to the previously described by applying pressure to port "B."



General Dimensions:



Model	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	A120	A180	Air connection
TM 032DA	110	45	45	65	22.5	23	12	50	25		F03 Ø36		M5X5	9				G 1/8"
TM 052DA	143	72	55	92	30	41	14	80	30	F05 Ø50	F03 Ø36	M6X8	M5X6	11	Ø40	158	200	G 1/4"
TM 063DA	190	86	69	106	35	45	16	80	30	F07 Ø70	F05 Ø50	M6X13	M5X10	14	Ø40	164	233	G 1/4"
TM 075DA	207	99.5	100.5	119.5	38.5	52.5	20.5	80	30	F07 Ø70	F05 Ø50	M6X10	M5X8	14	Ø40	103	243	G 1/4"
TM 083DA	213	109	88	129	46	52.5	21	80	30	F07 Ø70	F05 Ø50	M6X13	M5X10	17	Ø40	221	280	G 1/4"
TM 092DA	258	117	98.5	137	50	61	21	80	30	F07 Ø70	F05 Ø50	M6X12	M5X10	17	Ø40	260	274	G 1/4"
TM 0125DA	340	155	120.5	175	67.5	70	27.5	80	30	F10 Ø102	F07 Ø70	M6X16	M5X13	22	Ø65	365	470	G 1/4"
TM 140DA	414	171.5	132	191.5	75	76	32	80	30	F12 Ø125	F10 Ø102	M6X20	M5X15	27	Ø65	442	568	G 1/4"

Performance Data:

Air Consumption				
Air volume opening and closing				
Model	Air volume opening (L)	Air volume closing (L)	Weight	
			DA	SR
TM 032	0.035	0.045	0.72	
TM 052	0.09	0.12	1.3	1.4
TM 063	0.14	0.2	1.9	2.9
TM 075	0.21	0.3	2.7	3.2
TM 083	0.29	0.41	3.1	3.6
TM 092	0.49	0.71	5.1	5.9
TM 125	0.9	1.5	9.4	10.5
TM 140	1.7	2.4	13.7	16.2

Operating Conditions:

Operating Media: Dry or lubricated air, or inertia gas, which should be compatible with inside parts and lubricant of the cylinder.

Operating Temperatures: -20°C to 80°C (-4°F to 176°F) for Buna N (Std).
 -20°C to 120°C (-4°F to 250°F) for Viton (optional).
 -40°C to 80°C (-4°F to 176°F) for EPDM (optional).

Pressure Range for Double Acting (DA): 2bar to 8bar (30 to 116 PSI)

Pressure Range for Spring Return (SR): 3bar to 8bar (44 to 116 PSI)

Torque Ratings for Double Acting Actuators (Unit: Nm):

OPERATING PRESSURE (UNIT:BAR)							
Model	2	3	4	5	6	7	8
TM 032DA	2.78	4.20	6.00	7.50	9.00	10.00	11.50
TM 052DA	8.32	12.48	16.64	20.8	24.96	29.12	33.28
TM 063DA	14.64	21.96	29.28	36.6	43.92	51.24	58.56
TM 075DA	23.5	35.3	47	58.8	70.5	82.3	94
TM 083DA	29.7	44.5	59.4	74.2	89.1	103.9	118.8
TM 092DA	45.5	68.2	91.1	113.7	136.4	159.2	181.9
TM 0125DA	116.6	174.9	233.2	291.5	349.8	408.1	466.4
TM 140DA	175.48	263.22	350.96	438.7	526.44	614.18	701.92

Torque Ratings for Single Acting Actuators:

OUTPUT TORQUE OF AIR SUPPLY												Spring Stroke	
Pressure of a supply (bar)		3		4		5		6		7			
Model	Spring Qty	0	90	0	90	0	90	0	90	0	90	Start	End
TM052SR	5	8.48	6.28	12.64	10.44							4	6.2
	6	7.68	4.98	11.84	9.14							4.8	7.5
	7	6.98	3.78	11.14	7.94							5.5	8.7
	8			10.34	6.74	14.5	10.9					6.3	9.9
	9			9.54	5.44	13.7	9.6					7.1	11.2
	10			8.74	4.24	12.9	8.4	17.06	12.56			7.9	12.4
	11					12.1	7.1	16.26	11.26	20.42	15.42	8.7	13.7
TM063SR	12					11.3	5.9	15.46	10.06	19.62	14.22	9.5	14.9
	5	15	11.2	22.3	18.5	29.6	25.8					7	10.8
	6	13.5	9	20.8	16.3	28.1	23.7					8.5	12.95
	7	12	6.9	19.4	14.2	26.7	21.5					9.9	15.1
	8			18	12	25.3	19.3	32.6	26.6			11.3	17.3
	9			16.5	9.9	23.9	17.2	31.2	24.52			12.7	19.4
	10			15.3	7.7	22.6	15	29.9	22.3	37.2	29.6	14	21.6
TM075SR	11			13.8	5.6	21.1	12.9	28.4	20.2	35.7	27.5	15.5	23.7
	12					19.7	10.7	27	18	34.3	25.3	16.9	25.9
	5	23.4	17.8	35.1	29.5							11.9	17.5
	6	21.1	14.3	32.8	26							14.2	21
	7	18.7	10.8	30.4	22.5							16.6	24.5
	8			28	19	39.8	30.8					19	28
	9			25.7	15.5	37.5	27.3					21.3	31.5
TM083SR	10			23.3	12	35.1	23.8	46.8	35.5	58.6	47.3	23.7	35
	11					32.7	20.3	44.4	32	56.2	43.8	26.1	38.5
	12					30.4	16.8	42.1	28.5	53.9	40.3	28.4	42
	5	30.9	23.8	46.1	38.9							14.5	21.7
	6	28.1	19.5	43.3	34.6							17.39	26
	7	25.2	15.1	40.3	30.2							20.3	30.4
	8			37.4	25.9	52.6	41.1					23.2	34.7
TM092SR	9			34.5	21.5	49.7	36.7					26.1	39.1
	10			31.6	17.2	46.8	32.4	62	47.6	77.1	62.7	29	43.4
	11					43.9	28.1	59.1	43.3	74.2	58.4	31.9	47.7
	12					41	23.7	56.2	38.8	71.3	54	34.78	52.08
	5	50.28	37.78	75.54	63.03							25.5	38
	6	45.18	30.18	70.44	55.44							30.6	45.6
	7	40.08	22.58	65.34	47.84							35.7	53.2
TM0125SR	8			60.24	40.24	85.5	65.5					40.8	60.8
	9			55.14	32.69	80.4	57.9					45.9	68.4
	10			50.04	25.04	75.3	50.3	100.56	75.56	125.82	100.82	51	76
	11					70.2	42.7	95.46	67.96	120.72	93.22	56.1	83.6
	12					65.1	35.1	90.36	60.36	115.6	85.6	61.2	91.2
	5	115.5	88	173.8	146.3							59.4	86.9
	6	103.6	70.6	161.9	128.9							71.3	104.3
TM0140SR	7	91.8	53.5	150.1	111.6							83.1	121.6
	8			138.2	94.2	196.5	152.5					95	139
	9			126.3	76.8	184.6	135.1					106.9	156.4
	10			114.4	59.4	172.7	117.7	231	176			118.8	173.8
	11					160.9	100.4	219.2	158.7	277.5	217	130.6	191.1
	12					149	83	207.3	141.3	265.6	199.6	142.5	208.5
	5	174.7	131.2	262.5	219							88.5	132
TM0140SR	6	157	104.8	244.8	192.6							106.2	158.4
	7	133.9	78.4	227.1	166.2							123.9	184.8
	8			209.4	139.8	297.1	227.5					141.6	211.2
	9			191.7	113.4	279.4	201.1					159.3	237.6
	10			174	87	261.7	174.7	349.4	262.4	437.8	350.1	177	264
	11					244	148.3	331.7	236	419.5	323.8	194.7	290.4
	12					226.3	121.9	314	209.6	401.8	297.4	212.4	316.8

Numerous accessories are available as optional components to interface with the **TORQUEMATE 90** series actuators. Mounting kits also available.

Limit Switch Box with Optical Indicator:



This device is particularly suitable for monitoring the position of each valve, even at long distances.

It is manufactured according to VDI/VE 3845 STANDARDS. It contains two signaling limit switches either mechanical "SPDT", or proximity ones.

Spring loaded Cam operation
Sensing elements like micro switches SPDT
Proximity sensors
Enclosure in weather proof or exproof Gr.iiC
NAMUR mounting as standard
Special versions on request

Pneumatic Positioner



Pneumatic Positioner is particularly suitable for proportional working of both DAAND SR actuators.

This positioner is connected with the actuator with the means of NAMUR pad available on the output shaft on the positioner.

Supply : dry instrument air 2 to 7 bar.
Regulating signal pressure: from 3 to 15 psi
Protection : IP 55
Sensitivity : < 0, 5%range
Working temp. : from -20° C to +80° C

Electro Pneumatic Positioner output 4-20 mA
Position feedback transmitter also available.
ATEX version available on request.

Dec clutchable Gear Overrides:



VALVE MANIFOLDS



Nominal flow : 400 l/min
Protection class : IP 50(encapsulated d-sub plug)
Electrical Connection : Multi pin +6 field bus protocols
Application level : modular bus solutions
VTS cocept : subbase principle
Switching time : 12ms to 22ms
Working pressure range : 3-8 bar (internal pilot)

Maximum 12 plates either 12 5/2 valves or 24 3/2 valves, 5/2 and 3/2 plates are interchangeable. LED display, manual override.

NAMUR Solenoid Valves:



High Switch Reliability

NAMUR Interface

Operating Temperature: 5~60°C

Operating Pressure: 21~115PSI

Pressure Resistance: 170PSL

Coil Enclosure: NEMA4, 4x, 7, IP 65 (DIN 41050) All Coil are CE Approved

Coil Voltage: DC 24V (10%) AC 110-220V, 50/60 Hz

Explosion-protected Construction Are Available

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